RFID SMART ATTENDANCE SYSTEM

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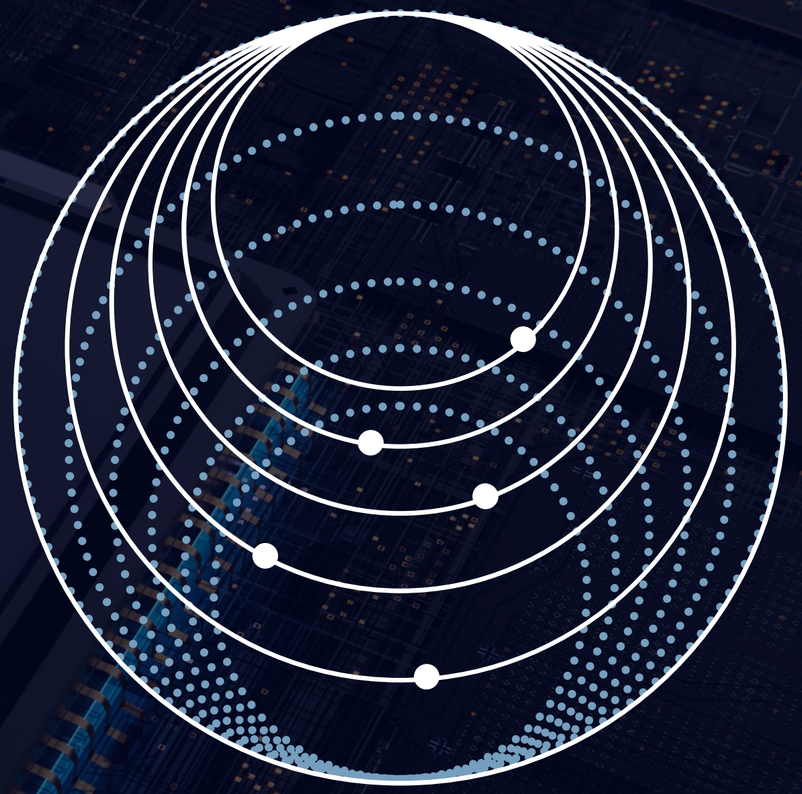
Problem Statement

“10 minutes lost per class = 60+ hours wasted per semester.”

- Manual attendance consumes 10–15 mins per lecture
- A typical college wastes 50–70 hours/semester on attendanceOver 30% institutions still use manual methods (education reports)

Pain:

- Lost teaching time
- Human errors & proxy attendance
- No real-time digital tracking



Solution & Features

Product Name: RFID SmartTap Attendance

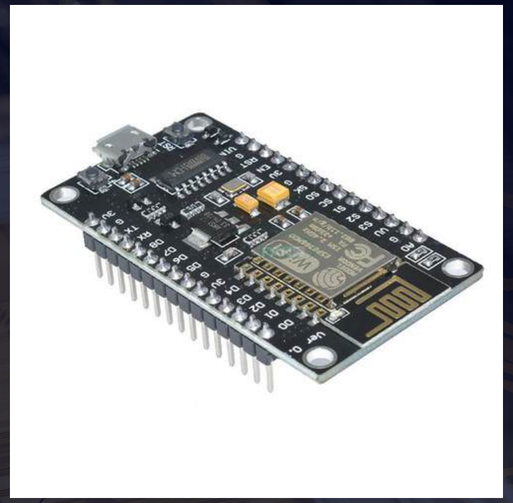
USP: Tap ID → Attendance Marked

→ Done in 1 second

Core Features:

- RFID Tap System → Instant attendance marking
- WiFi Sync (NodeMCU) → Real-time data upload
- Portable Device → Battery-powered usage
- LED Feedback → Instant success indication
- Digital Records → Easy tracking & storage

Tech Stack & Architecture



Tech Used:

- NodeMCU ESP8266 → WiFi communication & control
- RFID RC522 Module → Card detection & UID reading
- Li-Po Battery (3.7V, 2200mAh) → Portable power supply
- Charging Module (TP4056) → Safe battery charging
- Arduino IDE (Embedded C) → Programming

Why This Stack?

- Low-cost + scalable
- Easy IoT integration
- Real-time connectivity
- Architecture:
- RFID Card → RC522 → NodeMCU → WiFi → Database → Dashboard

Project Flow

Start → Student taps card → System reads UID → Data sent → Attendance marked → LED glows

Flow Breakdown:

1. Student taps RFID card
2. System reads unique ID
3. NodeMCU processes data
4. Sends to database via Wi-Fi
5. Attendance recorded instantly
6. LED confirms success

Outcome:

Fast, error-free attendance

Market Research & Competitive Edge

Market Insight:

- Smart education market growing rapidly (EdTech boom)
- Increasing demand for automation in institutions

Competitors:

- Biometric systems
- Manual registers
- Mobile-based attendance apps

Feature	Manual	Biometric	Our System
Speed	Slow	Medium	Fast
Cost	Low	High	Low
Portability	No	No	Yes
Accuracy	Low	High	High

Edge: Low-cost + portable + fast

Go-To-Market Strategy

Target Customers:

- Schools & Colleges
- Coaching Institutes
- Small offices

Business Model:

- B2B (sell to institutions)
- Device + setup cost
- Optional subscription for cloud dashboard

Expansion:

- Partner with colleges
- Bundle with smart classroom systems

Vision: Smart campuses everywhere

THANK YOU